Feature Engineering for Systematic Futures and Commodity Trading: An Institutional Research Primer

Executive Summary

Within systematic managed futures and commodity trading advisors (CTAs), the extraction of robust, economically viable, and statistically sound features from continuous price and return data represents the foundation of alpha generation.[1] While the availability of fundamental datasets, term structures, and physical supply chain metrics can enhance trading strategies, a vast and highly credible literature demonstrates that continuous futures prices and returns alone contain rich, tradeable information.[3] This primer is designed for quantitative researchers joining the systematic macro division to establish a rigorous framework for feature engineering. The primary challenge of trading a diverse universe of approximately 40 highly liquid contracts across energies, metals, agriculture, equity indexes, fixed income, and currencies is to construct features that are comparable across different assets while maintaining their unique physical and behavioral dynamics.[1] This report outlines a structured taxonomy of futures features, formalizes trend and momentum feature engineering with an emphasis on low-turnover estimators, details volatility and regime-detection frameworks, explores price-based proxies for commodity seasonality and physical inventory cycles, and establishes a blueprint for relative-value and cross-sectional features. Finally, the report defines best practices in feature selection, signal combination, and reproducible research pipeline design to ensure that all engineered features are institutionally credible and free from structural biases.[7]
The dataset under consideration spans from 2000 to 2026, comprising daily continuous open, high, low, close, volume, and open interest (OHLCV) bars across energy, precious metals, industrial metals, grains, softs, livestock, equity index futures, sovereign bonds, and foreign exchange contracts.[1] Given the structural constraint of having no direct access to individual contract histories, physical term structure curves, or inventory reports, the focus of this primer is the mathematical reconstruction of these physical state variables using high-efficiency estimators and higher-moment statistical signatures.[8] The objective is the design of robust, predictive features that can serve as clean inputs into a downstream portfolio optimization framework.[1]

Taxonomy of Futures Features

To build a coherent systematic trading framework, raw price and return series must be mapped into structured, risk-adjusted features.[1] The table below establishes the mathematical and economic taxonomy utilized across the institutional CTA space to categorize these feature classes.

Table 1: Taxonomy of Futures Feature Classes

Feature Class	Primary Mathematical Inputs	Key Economic / Behavioral Rationale	Target Asset Classes
Trend & Momentum	Continuous prices, log returns, OLS slope, t-statistics [5]	Capitalizes on investor under-reaction, anchoring, and risk-mitigation flows [13]	All asset classes (Equities, Fixed Income, FX, Commodities) [3]
Volatility & Regime	Daily Open, High, Low, Close (OHLC), rolling covariance matrices [8]	Captures risk-budgeting constraints, volatility clustering, and systemic liquidity shocks [1]	All asset classes (used primarily for risk scaling and filtering) [1]
Seasonality & Temporal	Calendar day, daily return signatures, trigonometric Fourier terms [20]	Models recurring physical supply/demand cycles, harvest windows, and heating/cooling demands [20]	Grains, Softs, Energies [20]
Price-Inferred Fundamentals	Realized volatility, rolling daily return skewness [9]	Proxies physical inventory levels, supply inelasticity, convenience yields, and hedging demand [9]	Commodities (Energies, Metals, Agriculture) [9]
Relative-Value / Cointegration	Pairwise log-price series, residuals of OLS regressions [27]	Exploits long-term economic relationships, substitution effects, and processing margins [27]	Highly linked commodity complexes (e.g., WTI vs. Brent, Corn vs. Soybeans) [27]

The distinction between convergent and divergent strategies represents a fundamental organizing principle in systematic macro.[13] Trend and momentum features are inherently divergent, aiming to capture episodic, uncharacteristic price expansions driven by behavioral biases and information dissemination lags.[13] Conversely, relative-value and cointegration-based features are convergent, identifying localized dislocations from long-term economic equilibria and trading the expectation of mean reversion.[13] Volatility and regime features do not generate directional alpha themselves, but function as critical risk-scaling overlays and state-dependent filters that modulate exposure based on market fragility.[1]

Trend and Momentum Feature Engineering

Mathematical Formulation of Trend Indicators

Traditional trend-following models utilize simple binary signs of past cumulative returns to determine position direction.[3] Under this paradigm, the standard Time Series Momentum (TSMOM) signal is formulated as:

$$X_t^{\mathrm{SIGN}} = \mathrm{sign}\left(\ln\left(\frac{C_t}{C_{t-L}}\right)\right)$$

where C_t is the continuous futures closing price at day t, and L is the lookback window (e.g., $L = 252$ trading days).[3] While robust over multi-decade horizons, this binary formulation induces significant portfolio turnover and transaction cost drag, especially during trend reversals or sideways, range-bound markets.[4]

To mitigate these transaction costs and improve the signal-to-noise ratio, institutional CTAs employ the *Time-Trend t-statistic (TREND)*.[4] Rather than evaluating returns between two arbitrary points, this method fits a linear trend to the past N-day daily futures price path using Ordinary Least Squares (OLS).[12] The underlying model is defined as:

$$F_\tau = \alpha + \beta\tau + e_\tau$$

where $\tau = 1, 2, \ldots, N$ represents the daily price steps within the lookback window, F_τ is the continuous daily futures price, β is the slope coefficient of the fit, and e_τ is the residual error term.[12]

The statistical significance of the estimated trend is quantified by the t-statistic of the slope coefficient, denoted as $t(\beta)$ [12]:

$$t(\beta) = \frac{\hat{\beta}}{SE\left(\hat{\beta}\right)}$$

To account for the high serial correlation and heteroscedasticity inherent in daily futures prices, the standard error $SE\left(\hat{\beta}\right)$ must be adjusted using the Newey-West variance-covariance estimator:

$$\hat{\Gamma} = \hat{\sigma}^2 \left(X'X\right)^{-1} \left(X'\Omega X\right) \left(X'X\right)^{-1}$$

where X is the design matrix containing the time index τ, and Ω is the heteroscedasticity and autocorrelation-consistent covariance matrix of the residuals.

Using this t-statistic, a systematic trading signal is constructed. The signal can be mapped in a binary format using critical threshold cutoffs [12]:

$$TREND_t = \begin{cases} +1 & \text{if } t(\beta) \geq +2 \\ -1 & \text{if } t(\beta) \leq -2 \\ 0 & \text{otherwise} \end{cases}$$

Alternatively, to maximize implementation efficiency, the signal can be mapped continuously to scale exposure dynamically between -1 and $+1$ based on the statistical significance of the trend [4]:

$$TREND_t^{\text{CONT}} = \text{clip}\left(\frac{t(\beta)}{2}, -1, 1\right)$$

By utilizing the statistical significance of the trend rather than simple return directions, the strategy abstains from taking positions when the trend is weak, thereby reducing portfolio turnover by approximately two-thirds and lowering execution costs by up to 35% without degrading gross performance.[4]

Multi-Horizon Trend Filters

A single trend speed exposes a CTA to significant style drift and timing risk.[6] Institutional portfolios combine trend features across a spectrum of timescales.[1] The table below delineates the characteristics of these different horizons.

Table 2: Comparative Analysis of Trend Horizons

Horizon	Lookback Window (N)	Reactive Speed	Drawdown Efficiency	Primary Behavioral Driver	Decelerating Mechanism
Fast	10 to 30 days [6]	High (captures rapid turning points) [6]	Low (prone to false positives/whipsaws) [6]	Market panic, stop-loss liquidations, FOMO [13]	Short-term mean reversion, microstructural decay [30]
Medium	60 to 125 days [6]	Moderate [6]	Moderate [6]	Information dissemination lags, commodity supply shocks [13]	Rolling consolidation, hedging flow absorption [9]
Slow	250 to 500 days [6]	Low [6]	High (acts as a stabilizing backbone) [6]	Multi-year macroeconomic and monetary policy regimes [13]	Macro regime shifts, long-term structural reversals [13]

Empirical analysis indicates that a robust managed futures program should not treat these horizons as separate strategies, but rather as a convex combination of shared building blocks, typically allocating risk across fast, medium, and slow sleeves to maximize the overall portfolio's Sharpe ratio.[6]

Volatility and Regime Features

High-Efficiency Volatility Estimators

To achieve stable risk targeting and perform volatility-adjusted return scaling, accurate estimates of ex-ante volatility are required.[1] While standard close-to-close historical standard deviation is widely used, it is statistically inefficient and discards approximately 93% of the intra-day price information contained in daily Open, High, Low, and Close (OHLC) bars.[8]

To address this, the *Yang-Zhang (YZ) Volatility Estimator* is implemented.[8] The YZ estimator is independent of drift and explicitly accounts for both overnight price gaps and open-to-close intra-day trends.[8] The total daily variance is formulated as [8]:

$$\sigma_{YZ}^2 = \sigma_o^2 + k\sigma_c^2 + (1-k)\sigma_{RS}^2$$

where the individual components are defined as follows:

1. **Overnight Variance (σ_o^2):** Measures the volatility of overnight price gaps using log open-to-close-of-prior-day returns [8]:

$$\sigma_o^2 = \frac{252}{n-1}\sum_{t=1}^{n}\left(\ln\left(\frac{O_t}{C_{t-1}}\right) - \bar{r}_o\right)^2$$

2. **Open-to-Close Variance (σ_c^2):** Measures the volatility of the daily trading session using log close-to-open returns [8]:

$$\sigma_c^2 = \frac{252}{n-1}\sum_{t=1}^{n}\left(\ln\left(\frac{C_t}{O_t}\right) - \bar{r}_c\right)^2$$

3. **Rogers-Satchell Variance (σ_{RS}^2):** A range-based estimator that captures volatility within the daily session while remaining unbiased in the presence of an arbitrary price drift [8]:

$$\sigma_{RS}^2 = \frac{252}{n}\sum_{t=1}^{n}\left[\ln\left(\frac{H_t}{C_t}\right)\ln\left(\frac{H_t}{O_t}\right) + \ln\left(\frac{L_t}{C_t}\right)\ln\left(\frac{L_t}{O_t}\right)\right]$$

4. **Weighting Factor (k):** Derived to minimize the statistical variance of the overall estimator for a given sample window n [8]:

$$k = \frac{0.34}{1.34 + \frac{n+1}{n-1}}$$

The practical implication of this mathematical formulation is significant: a 5-day Yang-Zhang volatility estimate achieves the same statistical precision as a 70-day standard close-to-close estimator.[8] Within a trend-following context, utilizing the YZ estimator for portfolio risk scaling reduces overall portfolio turnover by approximately 17% without degrading backtest performance.[5]

Macro-Regime Indicators and Systemic Stress

To dynamically scale risk or filter trend signals during periods of market stress, systematic macro strategies construct pricing-based regime indicators.[17]

Financial Turbulence Indicator

Based on the Mahalanobis distance, this indicator measures the "statistical unusualness" of a joint vector of asset returns.[16] It is formulated as [16]:

$$d_t = (y_t - \mu)\Sigma^{-1}(y_t - \mu)'$$

where y_t is a $1 \times M$ vector of daily log returns across the asset universe at time t, μ is the $1 \times M$ historical mean vector of returns, and Σ is the $M \times M$ historical covariance matrix.[16] High turbulence values (d_t) indicate that asset returns are either exhibiting extreme moves, or are violating historical correlation structures (e.g., typically correlated assets decoupling, or uncorrelated assets moving together in a liquidity sell-off).[16]

Systemic Risk Indicator (Gini of Eigenvalues)

Rather than using the standard Absorption Ratio, which requires an arbitrary, hard-coded choice of the number of principal components to include in the numerator, systematic pipelines implement the Gini coefficient of the eigenvalues of the rolling covariance matrix.[18]

Let the covariance matrix Σ, calculated over a rolling 250-week window, be decomposed into its eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_M$.[18] The Gini coefficient of these eigenvalues is calculated as:

$$G = \frac{\sum_{i=1}^{M} \sum_{j=1}^{M} |\lambda_i - \lambda_j|}{2M \sum_{i=1}^{M} \lambda_i}$$

A high Gini coefficient indicates that a small number of eigenvalues explain the vast majority of the variance across the futures universe.[18] Under this regime, the system is "tightly coupled," meaning that localized shocks are highly prone to propagating rapidly across different asset classes.[18]

Commodity-Specific Feature Engineering

Seasonality Features via Harmonic Regression

Agricultural commodities (Grains, Softs, Livestock) and energies exhibit strong seasonality driven by natural harvesting patterns and seasonal weather-driven demand.[20] However, using simple monthly dummy variables leads to overfitting and fails to capture the smooth, continuous transitions of physical production and consumption.[22]

To construct a robust seasonality feature, researchers fit a *Dynamic Harmonic Regression* to continuous log price series.[21] The seasonal component S_t is modeled as a linear combination of trigonometric Fourier terms [21]:

S_t = \sum_{k=1}^K \left

where K represents the number of Fourier sine/cosine pairs (harmonics).[21] A lower K (e.g., $K = 2$) produces a smooth seasonal curve, whereas a higher K (e.g., $K = 5$) allows the model to capture sharper seasonal spikes (e.g., natural gas spikes during winter heating demand or agricultural lows post-harvest).[20] The residual error of this regression is modeled using a non-seasonal $ARMA(p, q)$ process to capture short-term price autocorrelation and prevent spurious seasonal attribution.[22]

To stay ahead of the market, systematic strategies implement "proactive seasonality time-shifts".[23] Rather than placing trades concurrently with the seasonal peak, the seasonality feature is calculated with a one-month lead (front-running seasonality).[23] This allows the portfolio to establish positions before seasonal demand changes are fully priced in by commercial hedgers.[23]

Price-Inferred Physical Proxies under Data Constraints

When direct physical inventories and term structures cannot be accessed, daily price and return metrics can be engineered to proxy physical supply chain dynamics.[9] According to the theory of storage, physical supply constraints are highly asymmetric: while surplus inventory can be built up indefinitely, low physical inventory creates an absolute floor below which consumption must be rationed.[25] Consequently, when inventories are low, spot prices spike violently relative to forward prices, driving the market into backwardation, elevating short-term realized volatility, and generating highly positive daily return skewness.[9]

To capture this phenomenon without access to term structure curves or warehousing data, researchers construct the *Virtual Inventory Indicator (VII)*.[9] This indicator is formulated as a composite feature of rolling daily realized skewness and range-based volatility.[8] First, the rolling 12-month daily realized skewness is calculated as [9]:

$$Skew_t = \frac{\frac{1}{M-1}\sum_{\tau=t-M}^{t}\left(r_\tau - \bar{r}\right)^3}{\sigma_{CC}^3}$$

Next, the *Virtual Inventory Indicator* is defined as the product of cross-sectionally standardized skewness and short-term range-based volatility [8]:

$$VII_{i,t} = Z_{\text{cross}}\left(Skew_{i,t}\right) \times Z_{\text{cross}}\left(\sigma_{YZ,i,t}^{20D}\right)$$

where Z_{cross} denotes the cross-sectional z-score across the commodity universe on day t.[11] The physical and behavioral interpretation of this composite indicator is detailed below:

- **High Positive VII ($VII_{i,t} \gg 0$):** Indicates a tight inventory regime (backwardation).[39] Price spikes are frequent, spot volatility is elevated, and consumers are forced to pay a premium for immediate delivery.[40] Commodity spec flows and hedging demands are highly active.[9]
- **Negative or Neutral VII ($VII_{i,t} \leq 0$):** Indicates an abundant inventory regime (contango).[40] Price movements are quiet, volatility is compressed, and holding physical inventory incurs a carry drag.[40]

In institutional portfolios, this virtual inventory feature is utilized as a dynamic allocation overlay.[9] Under selective hedging models, commodities with high positive VII tend to be structurally overpriced by spec buyers chasing lottery-like upside payoffs, leading to lower subsequent returns.[9] Conversely, commodities with negative VII (high inventory buffers) are systematically underpriced, offering a highly robust risk-adjusted return profile.[9]

Relative-Value and Cross-Sectional Features

Cointegration-Based Futures Spreads

Relative-value strategies focus on convergent, mean-reverting anomalies rather than divergent trend behavior.[13] While correlation is a short-term relationship between stationary returns, cointegration describes a long-term economic tethering between non-stationary price series.[27] High correlation can be spurious, but cointegration ensures that the spread between two assets will revert to a constant mean.[27]

For a pair of continuous daily log-price series, $y_{1,t}$ and $y_{2,t}$ (e.g., WTI crude oil vs. Brent crude oil, or Corn vs. Soybeans), the Engle-Granger two-step cointegration model is formulated as [27]:

$$y_{1,t} = \alpha + \gamma y_{2,t} + \epsilon_t$$

where γ is the cointegration coefficient representing the hedge ratio, and ϵ_t is the residual spread.[27] The residual spread ϵ_t is tested for stationarity using the Augmented Dickey-Fuller (ADF) test.[28] If the null hypothesis of a unit root is rejected, the spread ϵ_t is stationary, confirming that the assets are cointegrated and the spread is mean-reverting.[27]

Once cointegration is established, the dynamics of the spread are modeled as a continuous-time Ornstein-Uhlenbeck (OU) process to extract trading signals [44]:

$$d\epsilon_t = \theta \left(\mu - \epsilon_t\right) dt + \sigma dW_t$$

where θ is the speed of mean reversion, μ is the long-term mean of the spread, σ is the volatility of the spread, and dW_t is a standard Wiener process.[44] The speed of mean reversion is used to calculate the half-life of deviations:

$$t_{1/2} = \frac{\ln(2)}{\theta}$$

This half-life serves as a quantitative framework for setting optimal holding periods and lookback windows.[44]

Fast Fourier Transform (FFT) for Lookback Optimization

To avoid arbitrary lookback parameter selection, researchers apply a discrete Fast Fourier Transform (FFT) to the spread series.[45] The FFT converts the time-domain spread into the frequency domain, isolating the dominant cyclical frequencies.[45] The half-wavelength of the primary frequency component is then dynamically selected as the rolling lookback window for the cointegration test, aligning the feature's sensitivity with the market's organic cyclical structure.[45]

Feature Engineering Used by Institutional CTAs

To demonstrate how leading systematic macro and managed futures firms implement price-based feature suites, the table below synthesizes institutional feature designs.

Table 3: Institutional Feature Implementations

Feature Name	Mathematical Proxy / Formulation	Institutional Users	Structural Benefit	Operational / Implementation Notes
Volatility-Scaled Returns	$r_t^{\text{adj}} =$ [1]	Man AHL, Aspect Capital, AQR [1]	Establishes a level playing field across diverse contract sizes and volatilities [1]	Requires a high-efficiency volatility estimator to minimize lagging and position whipsaws [5]
Trend t-statistic (TREND)	$t(\beta) =$ from rolling OLS price regressions [12]	Aspect Capital, Winton, Transtrend [13]	Reduces portfolio turnover by up to two-thirds relative to binary sign indicators [4]	Re-estimated daily using sliding windows from 20 to 500 trading days [12]
High-Efficiency Volatility	$\sigma^2_{\text{YZ}} = \sigma^2_o + k$ [8]	AQR, Man AHL, Systematica [46]	Maximizes information extraction from daily OHLC data; reduces noise in risk targeting [5]	Crucial for cross-sectional risk parity portfolios to prevent risk-budget overallocation [19]
Harmonic Seasonality	Fourier sine/cosine expansion with ARMA errors [21]	Winton, Man AHL [46]	Captures smooth, non-linear seasonal supply/demand patterns without overfitting [21]	Hyperparameter K (harmonics) must be tuned per sector using information criteria [22]
Cross-Sectional Z-Score	Rank-based normalization	GSA Capital, Quantica, AQR	Converts directional	Must be calculated

	of momentum scores across assets [11]	[3]	momentum into a market-neutral relative-value signal [19]	within sectors to avoid structural sector biases [11]

The historical trajectory of the CTA space indicates a steady evolution from simplistic trend filters to multi-feature, state-dependent architectures.[31] While pioneering models in the 1970s and 1980s focused purely on closing prices and moving average crossovers, modern quantitative researchers dedicate significant risk budgets to modulating factors—specifically volatility estimation, physical proxies, relative value, and seasonality systems.[31] This structural diversification is critical; over the 2010–2026 period, the standalone Sharpe ratios of simple trend-following systems have declined due to crowded execution, elevating the importance of low-correlation non-trend features.[32]

Feature Selection and Alpha Construction

Step 1: Multicollinearity Filtering

Before combining features into a composite alpha signal, multicollinearity must be addressed.[11] Many trend and momentum indicators across similar lookback windows are highly correlated, which can destabilize subsequent portfolio optimization.[6] To prevent this, pipelines implement the *Variance Inflation Factor (VIF)* method.[11]

For a set of P engineered features, each feature j is regressed against all remaining $P-1$ features [11]:

$$X_j = \beta_0 + \sum_{i \neq j} \beta_i X_i + e$$

The VIF for feature j is calculated as [11]:

$$VIF_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the coefficient of determination of the regression.[11] The feature with the highest VIF is iteratively dropped, and the process is repeated until all remaining features exhibit a VIF below a strict institutional threshold (typically $VIF < 10$).[11] This ensures a linearly independent basis of features across the portfolio.[11]

Step 2: Cross-Sectional Normalization & Sector Neutralization

To eliminate scale differences and prevent highly volatile sectors (such as Energy) from dominating the signal space, cross-sectional normalization is applied.[1]

1. **Winsorization:** Extreme outliers in the raw feature series $X_{i,t}$ are capped at the 1st and 99th percentiles within each daily cross-section to prevent distorted z-scores.[11]
2. **Z-Score Normalization:** The winsorized features are standardized within each sector s (e.g., Grains, Precious Metals, Financials) to ensure sector neutrality [11]:

$$Z_{i,t} = \frac{X_{i,t} - \mu_{s,t}}{\sigma_{s,t}}$$

where $\mu_{s,t}$ and $\sigma_{s,t}$ represent the cross-sectional mean and standard deviation of the feature within sector s at time t.[11] This normalization neutralizes structural sector tilts, converting absolute signal strengths into relative intra-sector rankings.[11]

Step 3: Akaike Information Criterion (AIC) Stepwise Optimization

To construct a predictive composite alpha without overfitting, the standardized, linearly independent features are selected using a multilinear predictive regression optimized for the *Akaike Information Criterion (AIC)*.[11] The regression models forward risk-adjusted returns:

$$\frac{r_{i,t+1}}{\sigma_{i,t}} = \beta_0 + \sum_{j=1}^{M} \beta_j Z_{i,j,t} + \epsilon_{i,t+1}$$

To identify the optimal feature subset, the pipeline runs a bidirectional stepwise OLS regression, evaluating the AIC at each step [11]:

$$AIC = 2M - 2\ln\left(\hat{L}\right)$$

where M is the number of active features and $\hat{L}$ is the maximum likelihood estimate of the model.[11] This penalty-based selection framework ensures that features are only added if their incremental predictive power outweighs the cost of increased model complexity.[11]

Step 4: Signal Combination Paradigms

Once the optimal subset of features is selected, they are combined into a tradeable composite signal using one of two primary institutional paradigms [10]:

1. **The Integrated Approach:** The individual standardized features are combined first (e.g., via a weighted linear combination) to form a single composite alpha signal.[10] Portfolio weights are then calculated based on this unified signal.[10]
2. **The Mixed Approach:** Separate sub-portfolios are constructed for each individual feature, and the resulting asset weights are then aggregated to form the final portfolio.[10]

In a long/short CTA framework with portfolio constraints (such as sector risk limits and leverage caps), the **Integrated Approach** is preferred.[10] Because the Mixed approach applies constraints to each sub-portfolio independently, it can strip out valuable cross-feature interaction terms, leading to information loss and sub-optimal risk-adjusted performance.[10]

Research Pipeline Design

Columnar Storage and Schema Design

To process daily OHLCV data across approximately 40 futures markets efficiently, systematic research pipelines utilize Apache Parquet as the primary storage format.[7] Parquet's columnar layout enables rapid query execution, efficient compression (via Snappy), and native integration with high-performance execution libraries such as Polars or Apache Arrow.[7] The table below defines the institutional schema design for storing continuous futures histories and engineered features.

Table 4: Institutional Parquet Schema Design

Field Name	Physical Type	Logical Type / Constraint	Description
ticker	BYTE_ARRAY	UTF8 / Primary Key	Unique contract identifier (e.g., "CL", "GC", "ES")
date	INT32	DATE / Index	ISO-8601 date index of the trading day
open	DOUBLE	Float64 / Non-nullable	Daily continuous open price (adjusted for roll gaps)

high	DOUBLE	Float64 / Non-nullable	Daily continuous high price (adjusted for roll gaps)
low	DOUBLE	Float64 / Non-nullable	Daily continuous low price (adjusted for roll gaps)
close	DOUBLE	Float64 / Non-nullable	Daily continuous close price (adjusted for roll gaps)
volume	INT64	Int64 / Nullable	Aggregate daily trading volume
open_interest	INT64	Int64 / Nullable	Aggregate end-of-day open interest
is_test_period	BOOLEAN	Bool / Non-nullable	Flag indicating out-of-sample test period [7]

Preventing Lookahead Bias

Lookahead bias is the primary failure mode in backtesting and must be systematically prevented during feature engineering.[7]
To enforce strict historical causal boundaries, systematic pipelines run automated unit tests [7]:

1. The signal generation codebase is executed on the full dataset, generating a matrix of historical weights W^{Full}.[7]
2. The pipeline is then executed on a truncated dataset (e.g., with all data after a specific date T_{trunc} removed), generating weights W^{Trunc}.[7]
3. The test asserts that:

$$W_t^{\text{Full}} = W_t^{\text{Trunc}} \quad \forall\, t \leq T_{\text{trunc}}$$

If any differences are detected beyond a strict floating-point tolerance (relative tolerance of

10^{-5}, absolute tolerance of 10^{-8}), the pipeline is flagged for lookahead bias and rejected.[7]

Determinism and Reproducibility

To ensure that backtest results are robust and reproducible across local research environments and production trading engines, the pipeline enforces strict computational constraints [7]:

- **Explicit Random Seeds:** Any stochastic operations (e.g., Monte Carlo simulations in AIC optimization or GARCH parameter initializations) must set explicit seeds at the main entry point.[7]
- **Ordered Iterations:** The pipeline avoids iterating over unordered sets or dictionaries, as different memory allocations can lead to non-deterministic calculations.[7] All iterations are performed over sorted index arrays.[7]
- **Dependency Locking:** Package environments are locked using precise dependency managers (such as uv in Python or renv in R) to guarantee identical mathematical execution across servers.[7]

Practical Recommendations

For a quantitative researcher building an institutional-quality CTA and systematic commodity trading portfolio, the following structural guidelines are recommended:

1. **Default to the Yang-Zhang Volatility Estimator:** Replace traditional close-to-close standard deviation with the Yang-Zhang estimator for risk budgeting and position sizing.[5] This reduces estimation noise and portfolio turnover by ~17%.[5]
2. **Transition from Sign to Trend t-statistics:** Rather than using the sign of past returns, fit rolling OLS price regressions and utilize the t-statistic of the slope ($t(\beta)$) to generate trend signals.[12] This transition smooths signal scaling and reduces transaction costs by up to 35%.[4]
3. **Model Seasonality via Fourier Terms:** Avoid simple monthly dummy variables for commodity seasonality.[22] Implement dynamic harmonic regressions with a low number of harmonics ($K \leq 3$) to capture clean, seasonal curves without overfitting.[21] Use a one-month phase shift to front-run seasonal positioning.[23]
4. **Leverage Realized Skewness as an Inventory Proxy:** In the absence of physical inventory data, calculate rolling 12-month realized daily return skewness.[9] Use this higher-moment signature as a proxy for convenience yields and supply inelasticity, structuring long-short alpha models that exploit skewness preferences and selective hedging flows.[9]
5. **Enforce Cointegration via FFT Windowing:** When constructing relative-value spreads, perform Engle-Granger cointegration tests on log prices rather than relying on correlation.[27] Apply Fast Fourier Transforms to dynamically set the rolling lookback window to the half-wavelength of the dominant cycle, ensuring the model adapts to the organic co-movement of the underlying assets.[45]

6. **Implement Automated Lookahead Tests:** Embed truncated data replication checks directly into the continuous integration (CI) pipeline.[7] Any pull request containing new features must pass this automated check to ensure absolute historical causality before backtesting begins.[7]

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